

8.1.1 $w_0 = 0$

2. Independent increments

3. Stationary normal increments

$$w_T - w_S \sim N(0, t - S)$$

4. continuous paths

(b) $\text{Cov}(w_s, w_t) = E[w_s w_t] - E[w_s]E[w_t] = E[w_s w_t]$
 $w_t = w_s + (w_t - w_s)$

$$E[w_s (w_t - w_s)] = E[w_s]E[w_t - w_s] = 0$$

$$\text{Cov}(w_s, w_t) = \min(s, t)$$

(c) $E[Q^n] = \sum E[(w_{t_{k+1}} - w_{t_k})^2]$

$$E[Q^n] = \sum t_{k+1} - t_k = T - 0 = T$$

(d) $\Delta t_k = t_{k+1} - t_k$

$$\text{Var}(x^2) = 2\sigma^4$$

$$\text{Var}(Q^n) = \sum 2(\Delta t_k)^2 \leq 2m_0 \times |\Delta F_{\pi}| \sum \Delta t_k$$

$$= 2|\pi|T$$

as $\pi \rightarrow 0$, $\text{Var}(Q^n) \rightarrow 0$

since $E[Q^n] = T$

$$\Delta \text{Var}(Q^n) \rightarrow 0, Q^n \rightarrow \pi \sqrt{2}$$

$$a) df(t, x_t) = \left(\frac{df}{dt} + \mu(t, x_t) \frac{df}{dx} + \frac{1}{2} \sigma^2(t, x_t) \frac{d^2f}{dx^2} \right) dt + \sigma(t, x_t) \frac{df}{dx} dW_t$$

$$b) f(S_t) = \ln(S_t)$$

$$\frac{\partial f}{\partial t} = 0, \quad \frac{\partial f}{\partial S} = \frac{1}{S}, \quad \frac{\partial^2 f}{\partial S^2} = -\frac{1}{S^2}$$

$$d(\ln S_t) = \frac{1}{S_t} \ln S_t dt + \sigma S_t dW_t + \frac{1}{2} \left(-\frac{1}{S_t^2} \right)$$

$$\int d(\ln S_t) = \int \left(\mu - \frac{\sigma^2}{2} \right) dt + \sigma dW_t$$

$$\ln(S_t) - \ln(S_0) = \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t$$

$$S_t = S_0 \exp \left\{ \left(\mu - \frac{\sigma^2}{2} \right) t + \sigma W_t \right\}$$

$$c) \text{ Correct } F[S_t] = F[S_t] = S_0 e^{(\mu - \sigma^2/2)t} E[e^{\sigma W_t}]$$

$$E[e^{\sigma W_t}] = e^{\sigma^2 t/2}$$

$$E[S_t] = S_0 e^{(\mu - \sigma^2/2)t} e^{\sigma^2 t/2} = S_0 e^{\mu t}$$

$$\text{incorrect} \Rightarrow S_t = S_0 e^{\mu t + \sigma W_t}$$

$$F[S_{inc}] = S_0 e^{\mu t} E[e^{\sigma W_t}] = S_0 e^{(\mu + \sigma^2/2)t}$$

Q3. (a), $df_t = -\theta x_t$

• $df_t = \sigma$

• financial meaning $\Rightarrow$ drift pushes rate toward 0. If x_t is +, the drift becomes highly -ive. If x_t is -ive, drift is +.

(1)
$$X_{K+1} = X_K - \theta X_K \Delta t + \sigma \Delta W_K$$

$$\Delta W_K \sim N(0, \Delta t)$$

(c) As $t \rightarrow \infty$, effect of x_0 vanishes, and the probab x_t stabilize to $N(0, \frac{\sigma^2}{2\theta})$

(g) • Strong conv
 $\Rightarrow$ Measure pathwise acc.

• Weak conv,
 $\Rightarrow$ Measure $\Rightarrow$ distributional acc.